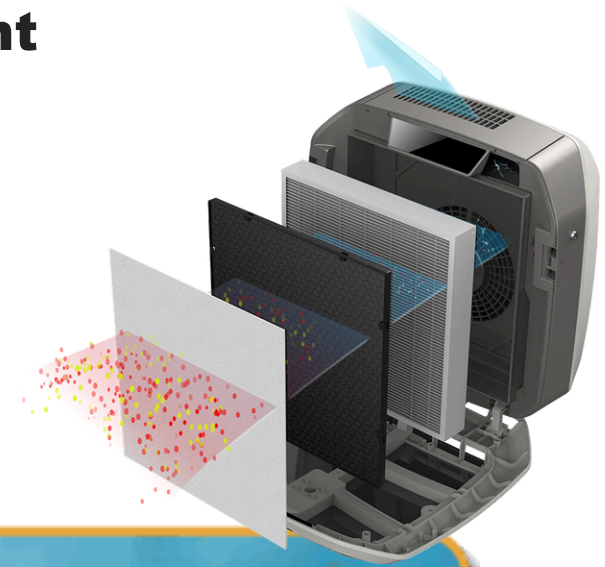


Market fit Research for Air Purifier development

Problem Statement Primary Analysis Secondary Analysis Critical Questions Asked



Problem Statement

"**Air Pure Innovations**" is a startup born out of the air quality crisis in India, with 14 cities ranking among the world's top 20 most polluted urban centers. The company is in the early stages of product development and is unsure whether there is a strong, sustained demand for its air purifier product. Before committing to production and R&D, they need to answer critical questions:

1. What pollutants or particles should their air purifier target?
2. What are the most essential features that should be incorporated into the air purifier?
3. Which cities have the highest demand for air purifiers, and what is the market size in these regions?
4. How can R&D be aligned with localized pollution patterns?

Dashboard

Problem Statement

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Market Fit Research for Air Purifier development

Primary Analysis

- 1. List the top 5 and bottom 5 areas with highest average AQI. (Consider areas which contains data from last 6 months: December 2024 to May 2025)
- 2. List out top 2 and bottom 2 prominent pollutants for each state of southern India. (Consider data post covid: 2022 onwards)
- 3. Does AQI improve on weekends vs weekdays in Indian metro cities (Delhi, Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad, Ahmedabad, Pune)? (Consider data from last 1 year)
- 4. Which months consistently show the worst air quality across Indian states — (Consider top 10 states with high distinct areas)
- 5. For the city of Bengaluru, how many days fell under each air quality category (e.g., Good, Moderate, Poor, etc.) between March and May 2025?
- 6. List the top two most reported disease illnesses in each state over the past three years, along with the corresponding average Air Quality Index (AQI) for that period.
- 7. List the top 5 states with high EV adoption and analyse if their average AQI is significantly better compared to states with lower EV adoption

Dashboard

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Market fit Research for Air Purifier development

Secondary Analysis

1. Which age group is most affected by air pollution-related health outcomes — and how does this vary by city?
2. Who are the major competitors in the Indian air purifier market, and what are their key differentiators (e.g., price, filtration stages, smart features)?
3. What is the relationship between a city's population size and its average AQI — do larger cities always suffer from worse air quality? (Consider 2024 population and AQI data for this)
4. How aware are Indian citizens of what AQI (Air Quality Index) means — and do they understand its health implications?
5. Which pollution control policies introduced by the Indian government in the past 5 years have had the most measurable impact on improving air quality — and how have these impacts varied across regions or cities?

Dashboard

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Market fit Research for Air Purifier development

Critical Questions asked

- Priority Cities: Which Tier 1/2 cities show irreversible AQI degradation?
- Health Burden: How do AQI spikes correlate with pediatric asthma admissions?
- Behavior Shifts: Do pollution emergencies increase purifier searches/purchases?
- Feature Gap: What do existing products lack (e.g., smart AQI syncing, compact designs)?

Dashboard

Problem Statement

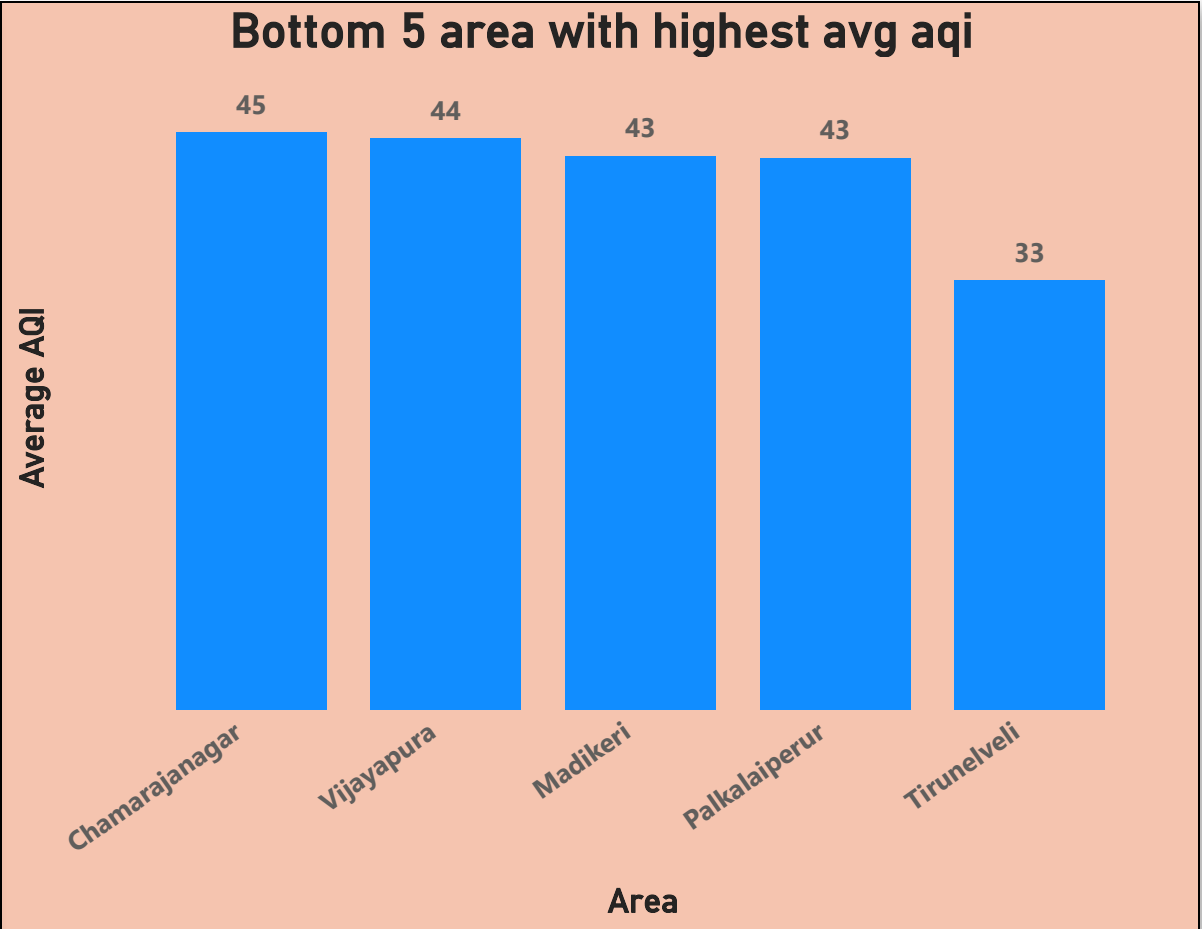
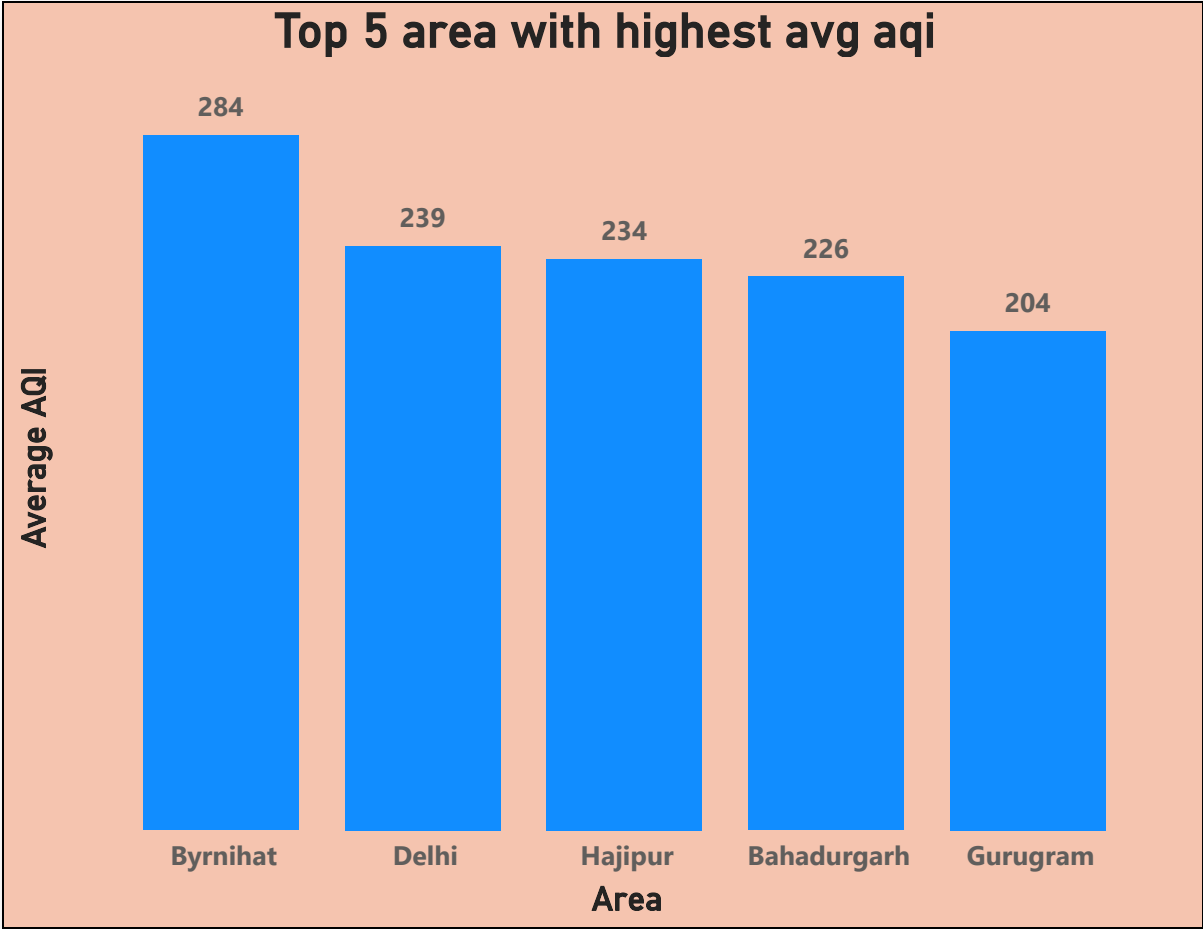
Primary Analysis

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Critical Questions Asked

1. List the top 5 and bottom 5 areas with highest average AQI.
(Consider areas which contains data from last 6 months: December 2024 to May 2025)

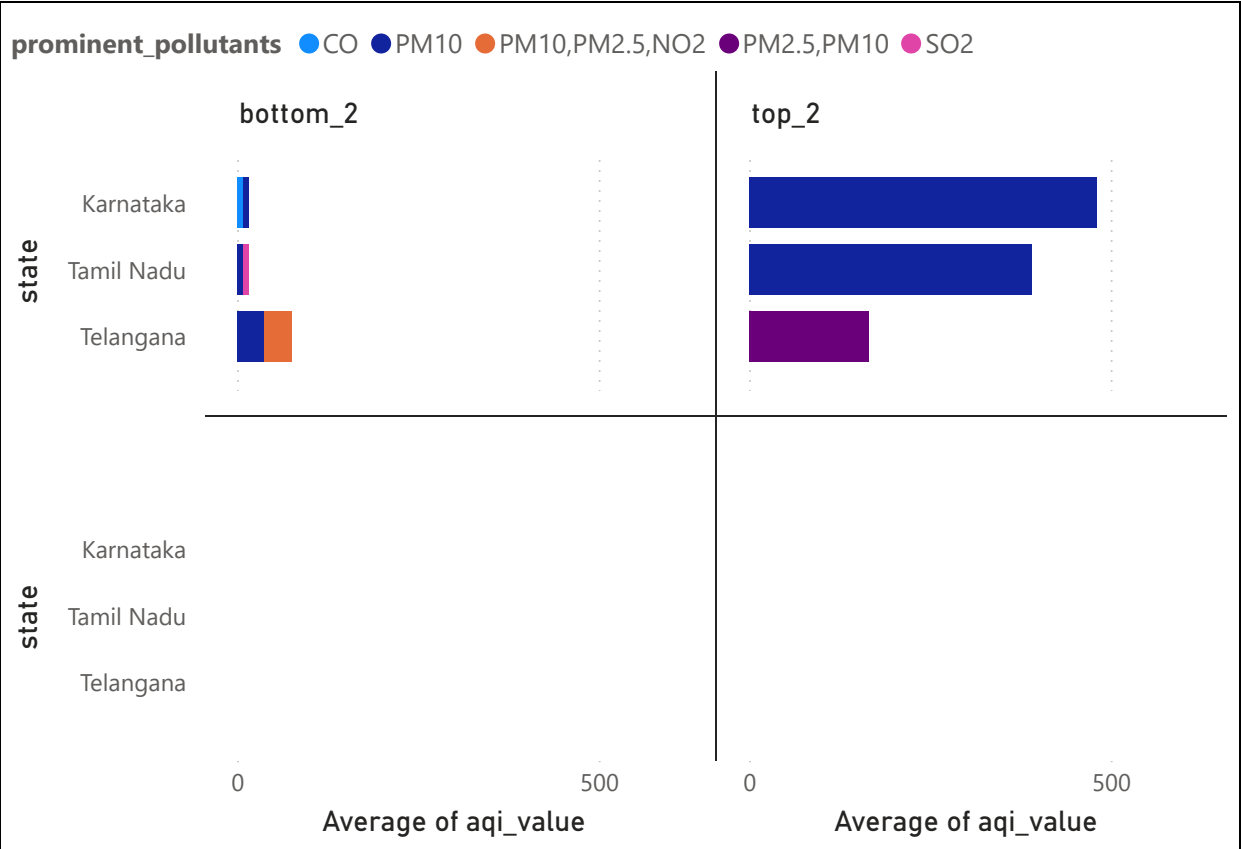
Dashboard
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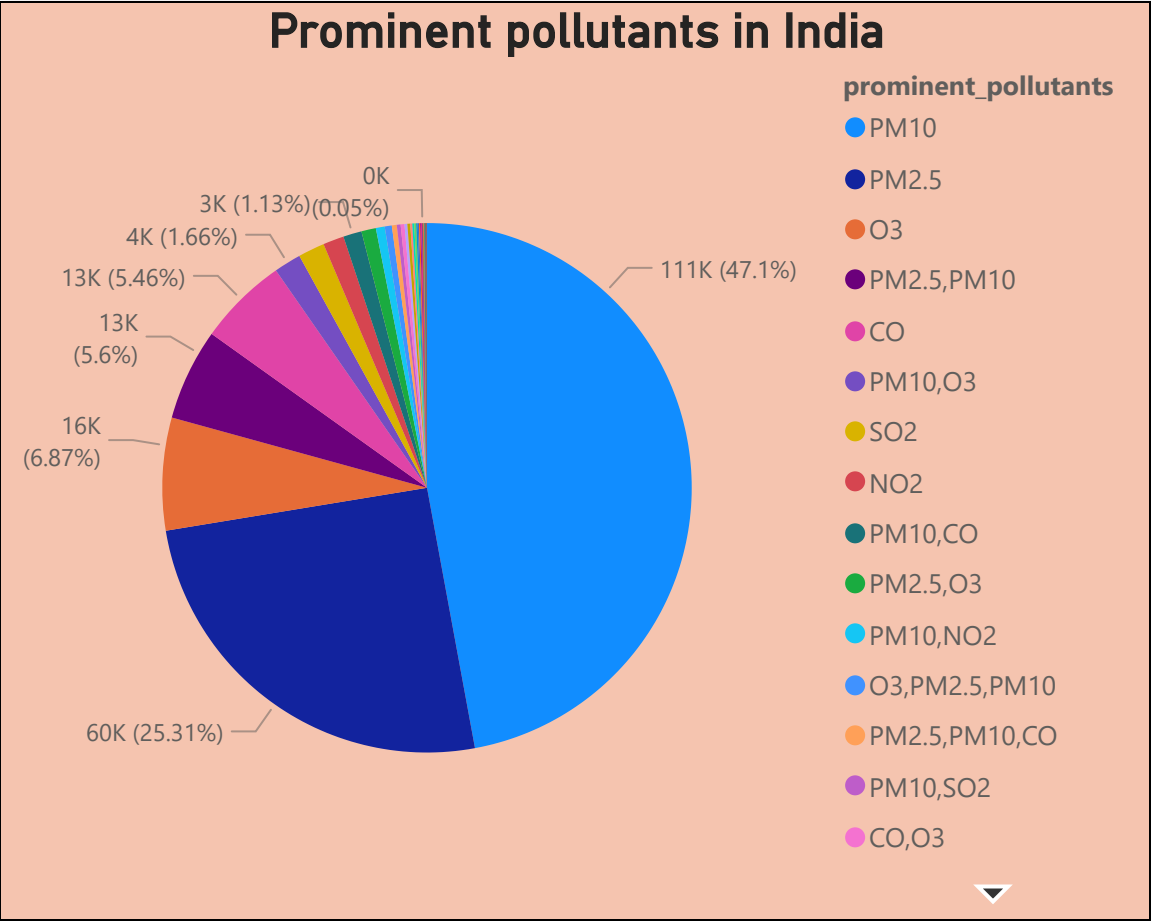
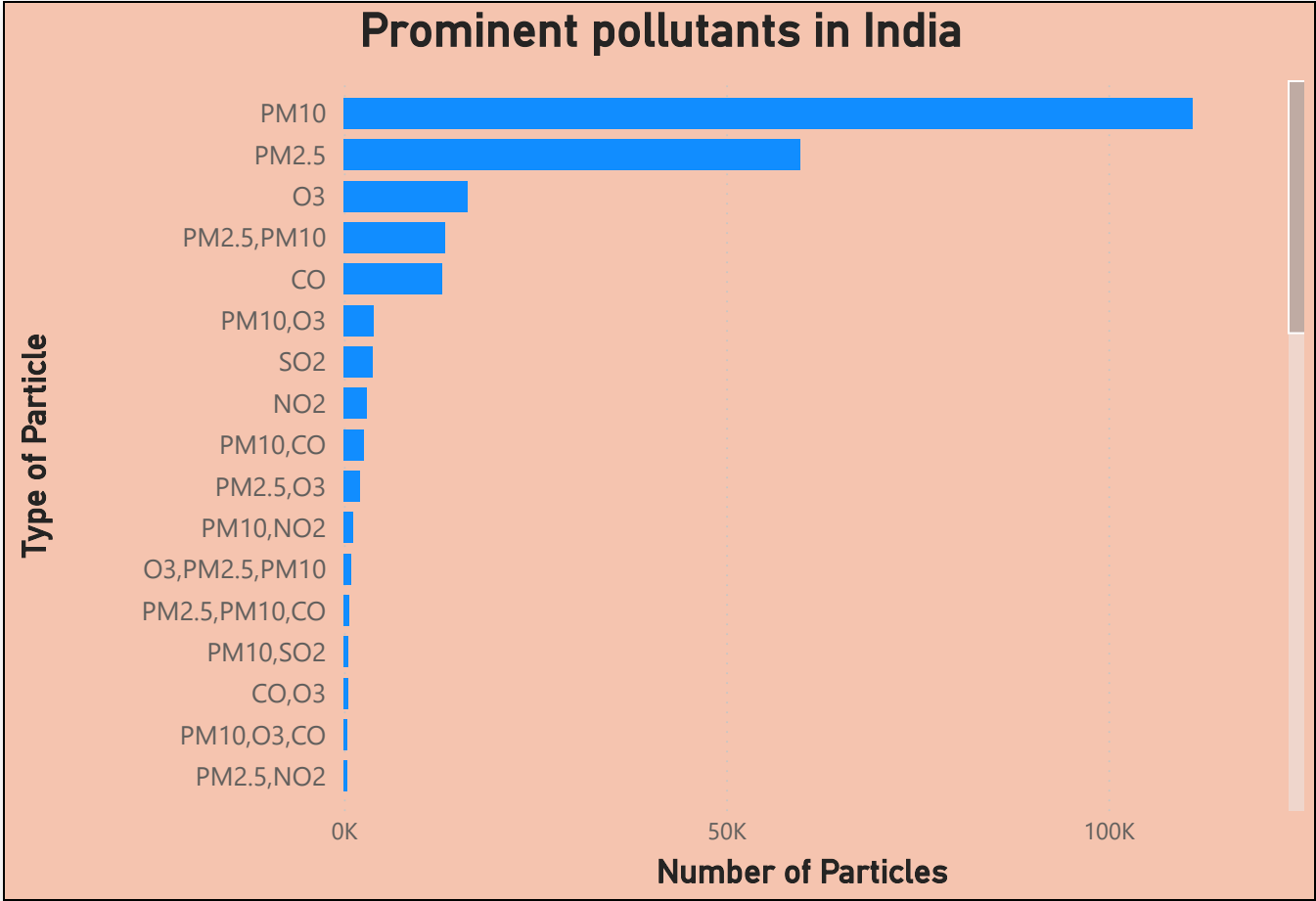
2. List out top 2 and bottom 2 prominent pollutants for each state of southern India.
(Consider data post covid: 2022 onwards)

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state	prominent_pollutants	Category	Average of aqi_value
Karnataka	CO	bottom_2	8.00
Karnataka	PM10	bottom_2	8.00
Karnataka	PM10	top_2	480.00
Tamil Nadu	PM10	bottom_2	8.00
Tamil Nadu	PM10	top_2	390.00
Tamil Nadu	SO2	bottom_2	8.00
Telangana	PM10	bottom_2	37.00
Telangana	PM10,PM2.5,NO2	bottom_2	38.00
Telangana	PM2.5,PM10	top_2	165.50
Total			168.15



Pollutant Particles to target



3. Does AQI improve on weekends vs weekdays in Indian metro cities (Delhi, Mumbai, Chennai, Kolkata, Bengaluru, Hyderabad, Ahmedabad, Pune)?
(Consider data from last 1 year)

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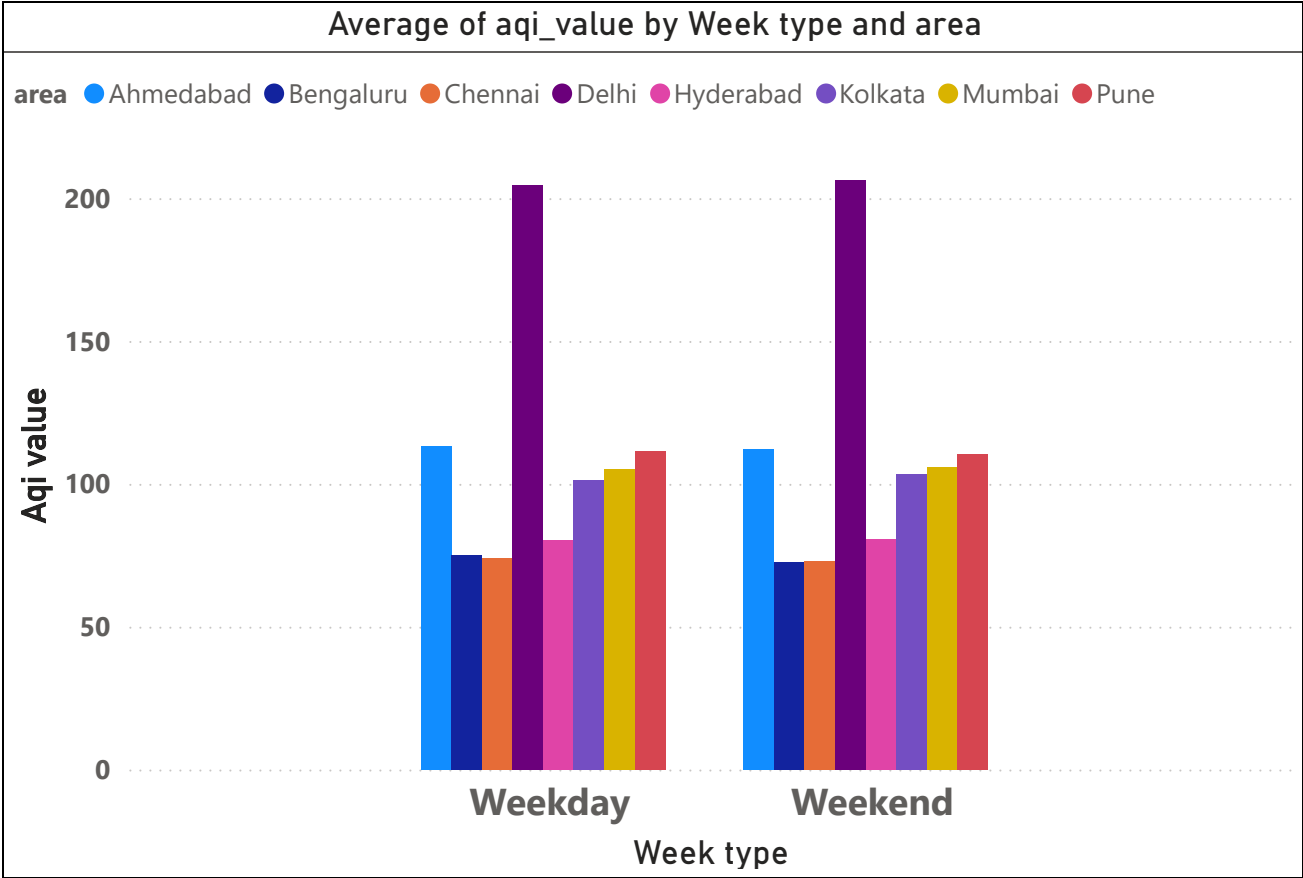
area

All

Week type

☐ Weekday

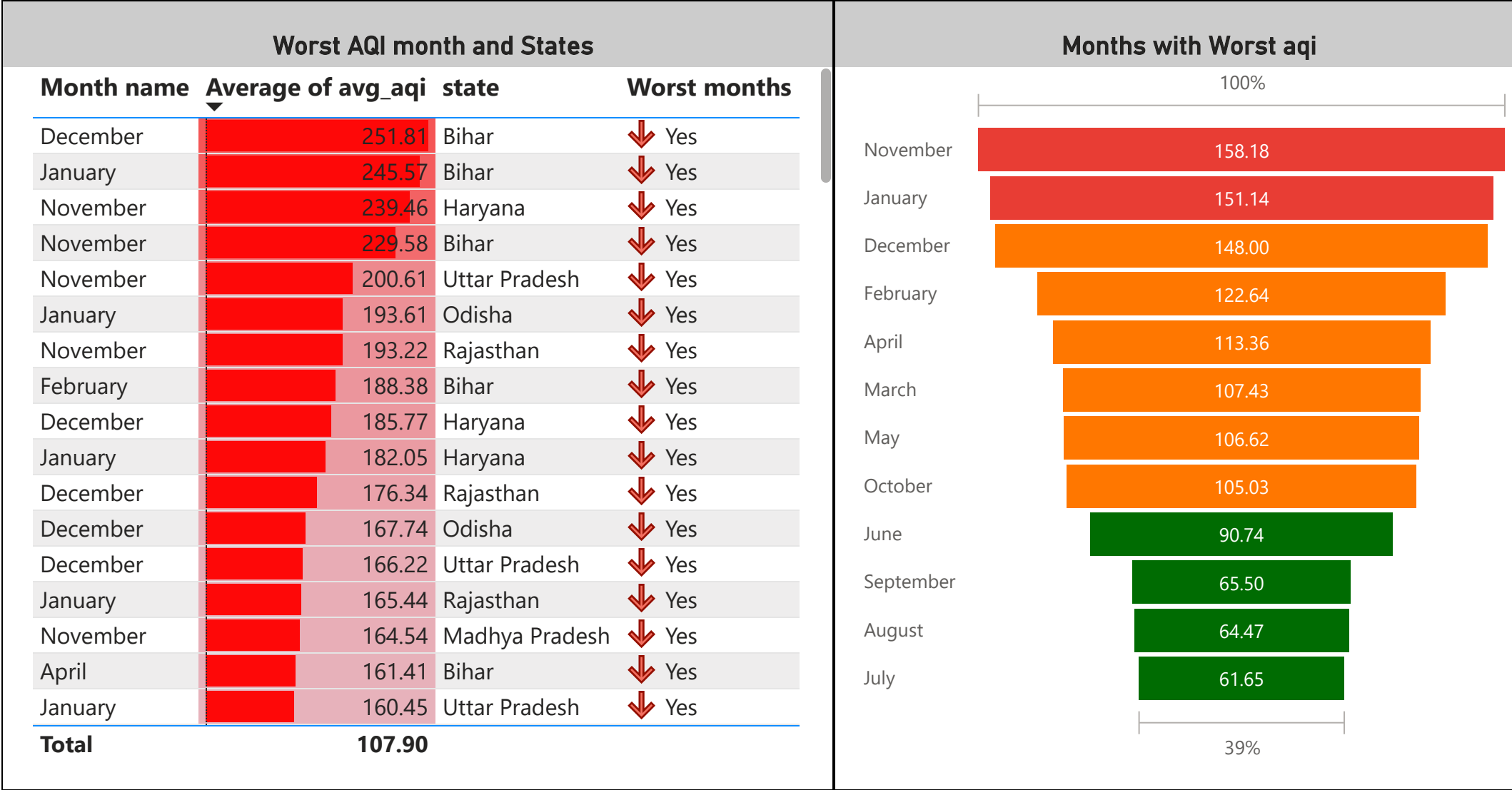
☐ Weekend



area	Weekday	Weekend	Total
Ahmedabad	113.14	112.21	113.60
Bengaluru	75.01	72.53	74.89
Chennai	74.07	72.90	74.01
Delhi	204.72	206.33	206.42
Hyderabad	80.36	80.80	81.01
Kolkata	101.46	103.34	102.03
Total	108.14	108.00	108.68

4. Which months consistently show the worst air quality across Indian states —
(Consider top 10 states with high distinct areas)

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Top 10 States with highest distinct areas
state
Uttar Pradesh
Tamil Nadu
Rajasthan
Odisha
Maharashtra
Madhya Pradesh
Karnataka
Haryana
Bihar
Andhra Pradesh

6. List the top two most reported disease illnesses in each state over the past three years, along with the corresponding average Air Quality Index (AQI) for that period.

Top two most reported disease illnesses in each state over the past three years, along with the corresponding average Air Quality Index (AQI) for that period

state	disease_illness_name	Average of aqi_value
Andaman and Nicobar Islands	Chickenpox	90.34
Andaman and Nicobar Islands	Fever with Rash	90.34
Andhra Pradesh	Scrub Typhus	96.56
Andhra Pradesh	Typhoid	96.56
Arunachal Pradesh	Suspected Mushroom Poisoning	96.68
Arunachal Pradesh	Typhoid	96.68
Assam	Typhoid	103.51
Bihar	Typhoid	97.91
Chhattisgarh	Typhoid	95.99
Dadra and Nagar Haveli	Fever	88.04
Dadra and Nagar Haveli	Fever of Unknown andigin (PUO)	88.04
Dadra and Nagar Haveli and Daman and Diu	Food Poisoning	108.65
Dadra and Nagar Haveli and Daman and Diu	Mumps	108.65
Delhi	Dengue	102.36
Delhi	Measles	102.36
Goa	Kyasanur Fandest Disease	98.68
Goa	Scrub Typhus	98.68
Gujarat	Typhoid	105.83
Total		111.13

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5. For the city of Bengaluru, how many days fell under each air quality category (e.g., Good, Moderate, Poor, etc.) between March and May 2025?

Bengaluru data between March and May 2025	
air_quality_status	Sum of Days
Moderate	13
Satisfactory	48
Total	61

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Top two most reported disease illnesses in each state over the past three years, along with the corresponding average Air Quality Index (AQI) for that period		
state	disease_illness_name	Average of aqi_value
Andaman and Nicobar Islands	Chickenpox	90.34
Andaman and Nicobar Islands	Fever with Rash	90.34
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Andhra Pradesh	Typhoid	96.56
Arunachal Pradesh	Suspected Mushroom Poisoning	96.68
Arunachal Pradesh	Typhoid	96.68
Assam	Typhoid	103.51
Bihar	Typhoid	97.91
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Goa	Kyasanur Fandest Disease	98.68
Goa	Scrub Typhus	98.68
Gujarat	Typhoid	105.83
Total		111.13

7. List the top 5 states with high EV adoption and analyze if their average AQI is significantly better compared to states with lower EV adoption.

Dashboard

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Primary Analysis

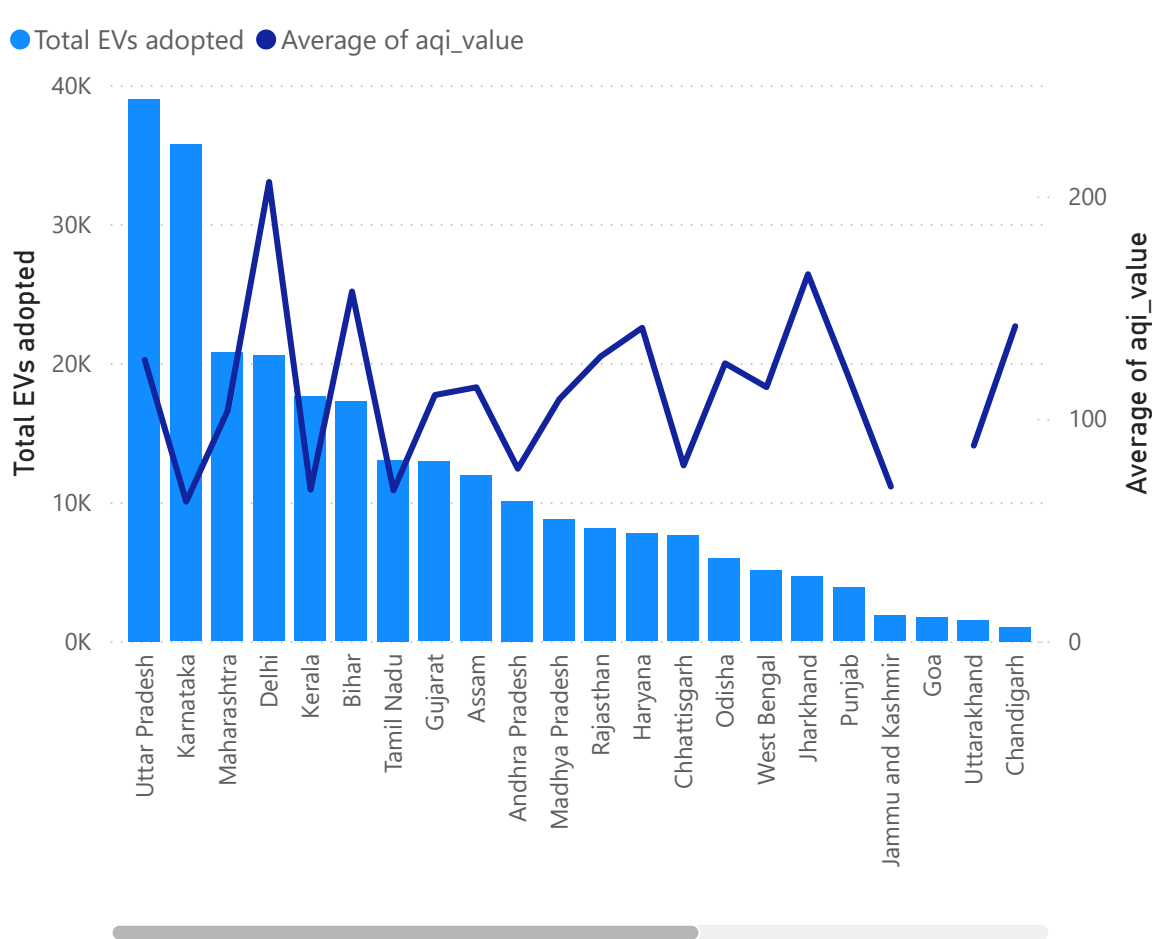
Secondary Analysis

Critical Questions Asked

Top 5 States with high EV adoption table

state	Total EVs adopted	Avg AQI	Population
Uttar Pradesh	38984	126.42	4526863
Karnataka	35717	62.67	3659615
Maharashtra	20755	103.64	4970780
Delhi	20530	206.42	2584182
Kerala	17605	68.07	2309511
Bihar	17251	157.16	2435691
Tamil Nadu	13029	67.76	3022227
Gujarat	12911	110.63	3416733
Assam	11891	114.12	2005415
Andhra Pradesh	10083	77.51	2413707
Madhya Pradesh	8738	108.72	1761225
Rajasthan	8082	127.99	2086122
Haryana	7735	140.85	1714946
Chhattisgarh	7613	78.99	1581966
Odisha	5952	124.88	1695372
West Bengal	5087	114.19	3246546
Total	258525	111.13	85248778

Top 5 states with high EV adoption bar chart



1. Which age group is most affected by air pollution-related health outcomes — and how does this vary by city?

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City	Most Affected Age Group	Major Health Outcomes	Key Reasons
Bengaluru	Adults (25–50)	Fatigue, respiratory irritation	Commuting exposure, tech workforce lifestyle
Delhi	Children (<14) & Elderly (>60)	Pediatric hospitalizations, elderly COPD & cardiac deaths	Highest PM2.5 levels, industrial + vehicular emissions
Kanpur	Children & Adults (15–45)	High under-5 mortality, asthma in working adults	Industrial emissions, biomass burning, poor circulation
Kolkata	Elderly (60+)	Stroke, cardiac mortality	Old vehicles, construction dust, high humidity
Lucknow	Children (<14)	Respiratory issues in children	Stagnant winter air, open waste burning
Chennai	Children (<14)	Allergic rhinitis, sinusitis, low birth weight	Thermal power, seasonal AQI spikes
Mumbai	Adults (30–55)	Heart disease, asthma	Traffic fumes, dense urban population

2. Who are the major competitors in the Indian air purifier market, and what are their key differentiators (e.g., price, filtration stages, smart features)?

Dashboard

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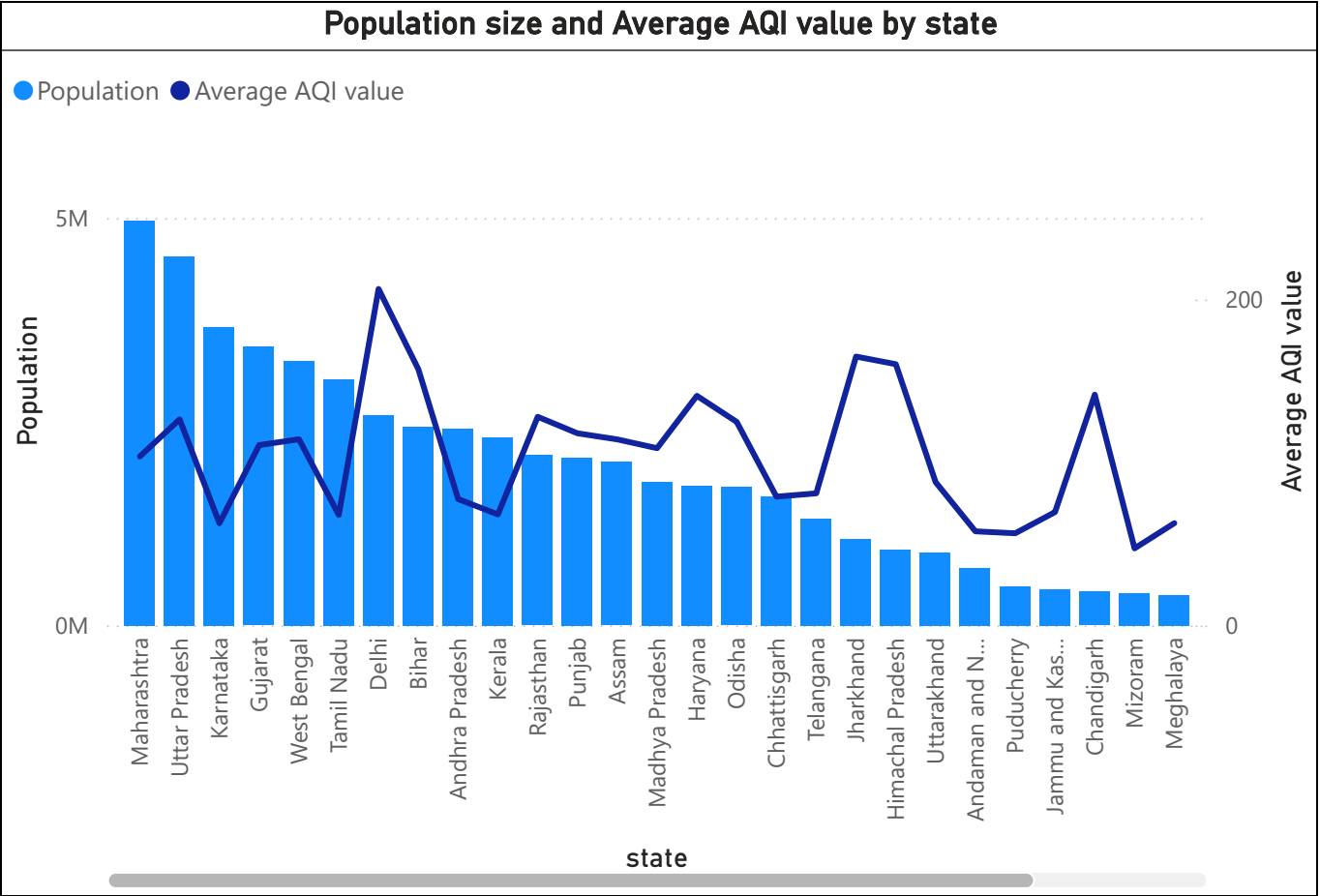
Critical Questions Asked

Brand	Filtration Technology	Key Differentiators	Smart Features	Price Range (INR) ▼
Philips	HEPA + VitaShield	VitaShield, trusted brand	App Control, Child Lock	₹9,000–₹25,000+
Coway	HEPA + Multi-stage	Long filter life, Korean tech	Low Noise, Auto Mode	₹9,000–₹25,000
Honeywell	HEPA + Carbon	Value-for-money, reliable	Auto Mode, No App Control	₹8,000–₹20,000
Kent	HEPA + UV + Ionizer	Budget ionizer combo	Basic Only	₹8,000–₹20,000
Mi (Xiaomi)	HEPA + Carbon + App Control	Smart features, affordable	App Control, Real-time AQI	₹7,000–₹15,000
Livpure	HEPA + Carbon	Entry-level, budget focused	Basic	₹6,000–₹12,000
Dyson	HEPA + Activated Carbon + Sensors	Premium design, high-end sensors	App Control, AQI Display, Auto Mode	₹35,000–₹60,000+
Blueair	HEPASilent + Particle Filter	Silent & stylish, premium build	App + AQI, Silent Operation	₹18,000–₹45,000
Samsung / LG	Multi-stage HEPA + Smart Sensors	Smart home integration	App Control, Voice Assistant	₹15,000–₹30,000
Sharp	HEPA + Plasmacluster Ion	Plasmacluster Ion tech, Japanese build	Basic, Energy Efficient	₹12,000–₹28,000

3. What is the relationship between a city’s population size and its average AQI — do larger cities always suffer from worse air quality? (Consider 2024 population and AQI data for this)

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Population size and average AQI by state			
state		Average of aqi_value	Population size
Nagaland	↑	80.46	82108
Manipur	↓	100.93	131938
Tripura	↓	126.70	316234
Sikkim	↑	53.69	352627
Arunachal Pradesh	↑	54.49	368633
Meghalaya	↑	62.73	368700
Mizoram	●	47.25	396244
Chandigarh	↓	141.56	414781
Jammu and Kashmir	↑	69.50	446124
Puducherry	↑	56.51	477480
Andaman and Nicobar Islands	↑	57.71	707064
Uttarakhand	↑	87.90	893142
Himachal Pradesh	⊗	160.26	932841
Jharkhand	⊗	164.94	1060804
Telangana	↑	81.01	1308093
Chhattisgarh	↑	78.99	1581966
Total		111.13	85248778



4. How aware are Indian citizens of what AQI (Air Quality Index) means — and do they understand its health implications?

Dashboard

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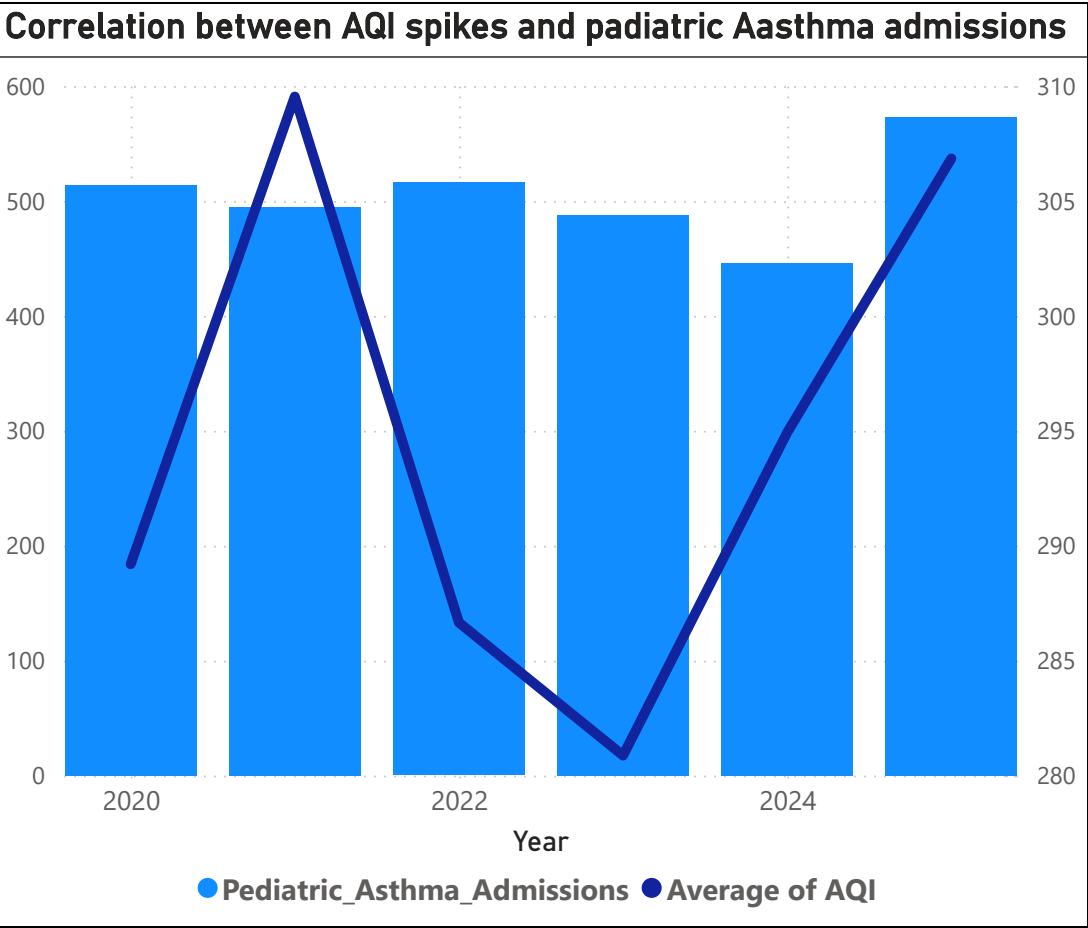
Data about Awareness about AQI among Indians (Perception Study on Air Quality in 17 Cities BY CMSR)			
Demographic Category	Sum of Aware & Understand AQI (%)	Sum of Aware but Don't Understand (%)	Sum of Not Aware & Don't Understand (%)
Govt. Employees	44.80	17.80	37.40
18–25	38.50	21.70	39.80
46–60	37.90	16.90	45.20
Private Sector	36.80	23.10	40.00
Students	36.50	17.00	46.40
Men	35.80	17.20	47.10
Below 18	35.10	14.30	50.60
Overall Respondents	35.00	19.10	45.90
Women	34.00	21.60	44.50
26–35	33.40	21.80	44.80
36–45	32.80	21.40	45.70
60+	32.70	10.70	56.50
Homemakers	29.30	23.60	47.20
Total	462.60	246.20	591.10

• **Priority Cities: Which Tier 1/2 cities show irreversible AQI degradation?**

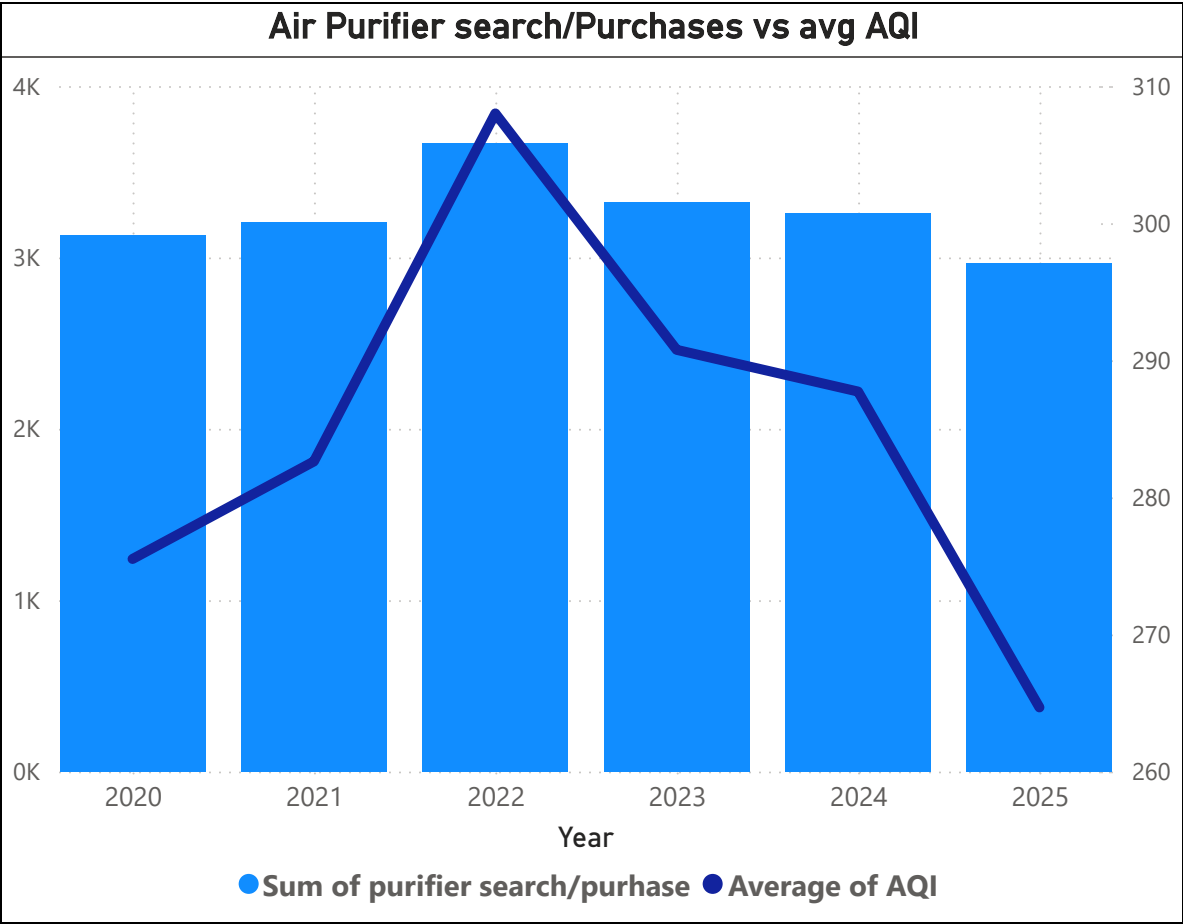
Cities with nearly irreversible AQI degradation				
City	Tier	AQI 2020	AQI 2025	Primary Pollutants
Kanpur	Tier 2	345	340	Industrial, PM2.5
Patna	Tier 2	330	325	Brick kilns, PM2.5
Ghaziabad	Tier 2	325	320	PM2.5, PM10
Delhi NCR	Tier 1	320	310	PM2.5, PM10, NOx
Agra	Tier 2	310	308	SO2, PM10
Faridabad	Tier 2	310	308	PM10, NO2
Lucknow	Tier 2	300	295	PM2.5, NO2
Varanasi	Tier 2	295	290	PM2.5, Biomass
Kolkata	Tier 1	290	285	PM2.5, SO2
Gwalior	Tier 2	280	278	Dust, PM10
Jodhpur	Tier 2	270	268	Dust, PM10
Amritsar	Tier 2	260	258	Biomass, PM2.5
Mumbai	Tier 1	250	245	PM2.5, NOx
Chennai	Tier 1	240	235	PM2.5, SO2
Total		4125	4065	

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Health Burden: How do AQI spikes correlate with pediatric asthma admissions?



Behavior Shifts: Do pollution emergencies increase purifier searches/purchases?



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Feature Gap: What do existing products lack (e.g., smart AQI syncing, compact designs)?

Dashboard

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Feature gap in existing Air Purifiers						
Feature / Product	Dyson	Mi	Atmotube Pro (Wearable AQI) ▼	AirVisual App (IQAir)	Plume Labs App	Opportunity
Predictive AQI Forecasts	✗	✗	⚠ via app only	✓	✓	Add to devices, not just apps
Health Recommendations	✗	✗	⚠ Manual	✓ (basic suggestions)	✓ (forecast tips)	Personalised, condition-based guidance
Multi-language Support	⚠ Partial	⚠ Limited	⚠ Limited	⚠ Limited	✓ Multilingual	Full multilingual + icon-based alerts
Smart AQI Sync (Gov data/API)	✗	✗	✓ via app	✓	✓	Add real-time syncing to purifiers and wearables
Personal Exposure Tracking	✗	✗	✓ Tracks exposure	⚠ City-based	✓ (limited)	Use GPS + time-location data for personal logs
Compact / Portable Design	✗	⚠ Medium	✓ Pocket-sized	✓ App only	✓ App only	Develop ultra-portable purifiers (for commuters etc.)
Community/Sharing Features	✗	✗	✗	✗	⚠ Some alerts	Add real-time local reports / alert sharing
Smart Home Integration (Alexa etc)	✓ Limited	✗	✗	✗	✗	Add IFTTT / Google Home / Alexa support

5. Which pollution control policies introduced by the Indian government in the past five years have had the most measurable impact on improving air quality — and how have these impacts varied across regions or cities?

Dashboard

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State	Sum of AQI 2020	Sum of AQI 2025	Improvement (%)	Schemes Implemented,
Uttar Pradesh	280	210	25.00	NCAP, Dust Control
Bihar	260	200	23.08	NCAP, LPG Expansion
Haryana	270	210	22.22	Stubble Management,
Gujarat	190	150	21.05	Dust Management,
Jharkhand	240	190	20.83	Mining Dust Control,
Chhattisgarh	200	160	20.00	Coal Power Regulation,
Punjab	250	200	20.00	Stubble Management,
Rajasthan	200	160	20.00	Urban Green Drive,
Madhya Pradesh	210	170	19.05	NCAP, Transport Reforms
West Bengal	220	180	18.18	Traffic Decongestion,
Telangana	170	140	17.65	EV Charging Infra,
Delhi	290	240	17.24	NCAP, Vehicle Ban
Odisha	190	160	15.79	NCAP, Port Emission Norms
Kerala	130	110	15.38	Indoor AQI Programs,
Assam	210	180	14.29	NCAP, Brick Kiln Shift
Tamil Nadu	140	120	14.29	NCAP, EV Push
Karnataka	150	130	13.33	NCAP, EV Subsidy
Maharashtra	180	160	11.11	BS-VI, Metro Expansion
Total	3780	3070	328.49	

How Research and development can be aligned with localized pollution patterns?

We can align Research and development with localized pollution patterns by targeting the most common polluting particles in a particular region and identifying the root cause of that particle concentration in that region. Then we can study health implications of these pollutant particles on people.

example

PM2.5 (Particulate Matter ≤ 2.5 micrometers) is the most harmful and dominant air pollutant in Delhi.

¹. These are **ultrafine particles** that penetrate deep into the lungs and even enter the bloodstream.

Consistently exceeds WHO safe limits (10–15 µg/m³ annual; Delhi often records 90–120+ µg/m³).

¹. It contributes to **cardio-respiratory illnesses, stunted lung growth in children**, and premature deaths.

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Particle_Type	Primary_Source	Approx_Contribution (%)	Seasonality_Impact
PM2.5	Agricultural Stubble Burning	25–35	Post-monsoon/winter
PM2.5	Biomass & Domestic Fuel Burning	12–24	High in winter
PM2.5	Industrial Emissions & Power Plants	11–18	Year-round
PM2.5	Road Dust & Re-suspended Soil	18–38	Year-round
PM2.5	Secondary Particles (NOx, SO2, VOCs)	Up to 30–33	High during winter
PM2.5	Vehicular Emissions	28–47	Year-round, peaks in winter
PM10	Construction & Demolition Activities	3–8	Year-round, mostly urban construction zones
PM10	Road Dust & Construction Debris	36–66	Year-round
PM10	Vehicular & Industrial Emissions	~10 each	Lower contribution to PM10 than PM2.5